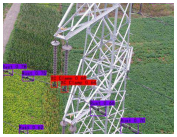
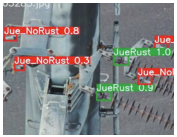
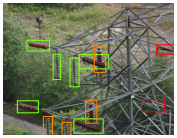
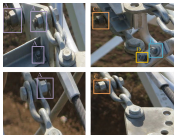
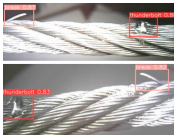
Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Defective Classification of PL Components		10,607 Images 1920 x 1080 x 3 39,150 patches	17 Classes 28,933 B. Box Num. 5 Classes w/ Defects	Open Available [244]	"Unsupervised" DL	DifferNet	AUC = 90.5%	[245]
					"Unsupervised" DL	AttentDifferNet	AUC = 94.34%	[246]
Rust Detection on PL Dampers and Clamps		3400 Images 416 x 416 x 3	4 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv7	mAP = 91.65%	[247]
					Supervised DL	RD-YOLO	mAP = 92.33%	[247]
Rust Detection on PL Dampers and Fittings		2705 Images 5184 x 3888 x 3	4 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv5	mAP = 90.15%	[248]
					Supervised DL	PWR-YOLOv5	mAP = 95.37%	[248]
PL Tower Health Condition Classification		Unspecified Images Num. 5616 x 3744 x 3 386,401 patches Unspecified Resol.	2 Classes Healthy Unhealthy	Not Available	Supervised DL	ResNet50v2	F1 = 84.13%	[249]
					Supervised DL	EfficientNetB0	F1 = 86.51%	[249]
Bolt Defect Detection on PL Fittings		1,852 Images Unspecified Resol.	6 Classes 14,635 B. Box Num.	Not Available	Supervised DL	Faster R-CNN	mAP = 47.2%%	[84]
					Supervised DL	UBDDM	mAP = 78.6%	[84]
PL Conductors Damage Detection		1,318 Images 1080 x 1080 x 3	2 Classes 1,512 B. Box Num.	Open Available [250]	Supervised DL	EfficientDet	mAP50 = 90.1%	[251]
					Supervised DL	Modified YOLOv8	mAP50 = 91.1%	[251]